$$h \in \mathbb{R}^{4096}$$

$\downarrow$ TPRization

$$u = \begin{bmatrix} u_{agent} \\ u_{patient} \\ u_{roles} \end{bmatrix} \qquad u \in \mathbb{R}^{3 \times 32}$$

$$u = A h + c$$

$\hookrightarrow A \in {}^{96 \times 4096}$

Suppose you want to change

$$\Delta u_{agent} = C_{prisoner} - C_{baby}$$

$$\Delta u = \begin{bmatrix} C_{prisoner} - C_{baby} \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta u = A \Delta h$$

$$\Delta h = A^{+} \Delta u = A^{T} (A A^{T})^{-1} \Delta u$$

(Moore-Penrose Pseudoinverse

$\hookrightarrow$ $\Delta h_1, \Delta h_2$ ... that satisfy $A \Delta h = \Delta u$

The pseudoinverse selects

$$\Delta h^{*} = \arg\min_{\Delta h} \|\Delta h\|_2 \text{ subject to } A \Delta h = \Delta u$$

$$A A^{+} \approx \mathbb{1}$$

$$h_{steered} = h + \alpha \Delta h$$